

Technology #174

Implement a cycle-to-cycle accumulation of viscosity estimates derived from measured flow rates, storing them in a passive lookup table that updates the P07 model's viscosity parameter before each new prediction.

Engineering Technology-Transfer Dossier
CONFIDENTIAL

Technology Maturity: ENGINEERING_DEFINITION

Generated: 2026-09-02

Not physically validated. Not transfer-ready.

BUYER DECISION PAGE

WHAT IS IT?

Implement a cycle-to-cycle accumulation of viscosity estimates derived from measured flow rates, storing them in a passive lookup table that updates the P-07 model's viscosity parameter before each new prediction.

WHY DOES IT MATTER?

Documented problem 'parameter estimation error' concerning viscosity model. User-submitted engineering problem (session input): "Can the P-07 model's viscosity parameter be corrected from measured flow? (model correction semantics)". [cod_optimade] retrieval status EMPTY — provider outcome recorded, not treated as absence (Art. XXI.3). [arxiv] 5 records retrieved live (2001-2025). Illustrative records: 'Modelling fluid deformable surfaces with an emphasis on biological interfaces'; 'Droplet break-up with negative momentum Fluid Dynamics Videos'. LIMITS STAMPED INLINE: literature context establishes the problem class, not event rates. Problem-existence basis: live records retrieved.

WHAT IS ACTUALLY ESTABLISHED?

Engineering definition with 3 governing equations, 13 design inputs, 6 failure modes. Computational model exists. No physical prototype.

WHAT IS NOT ESTABLISHED?

Physical validation, manufacturing process, regulatory pathway, IP ownership.

WHY MIGHT A BUYER CARE?

Opportunity in fluidics_hydraulic. 10-step build plan with specific test articles and acceptance criteria. Target buyer: domain-appropriate technology buyer (to be identified by portfolio owner).

WHAT DOES THE BUYER GET?

Buyer receives: canonical invention specification (with epistemic classes).; Buyer receives: engineering specification (this document, all blocks classed).; Buyer receives: design input/output graph with explicit IDs.; Buyer receives: engineering reasoning chains (claim->principle->model->input->assumption->output->failure->verification).; Buyer receives: killer-experiment definition and its EIG basis.; Buyer receives: evidence custody references and hashes.; Buyer receives: unresolved-uncertainty register.; Buyer receives: domain equation set (symbolic; sources and applicability judgments identified).

WHAT DOES THE BUYER HAVE TO BUILD?

Buyer must create: actual geometry and detailed design.; Buyer must create: sourced engineering parameters and acceptance thresholds.; Buyer must create: physical prototypes.; Buyer must create: all verification and validation results.; Buyer must create: manufacturing process qualification.; Buyer must create: regulatory pathway determination and submissions.

WHAT IS THE NEXT DECISIVE EXPERIMENT?

first physical article implementing the proposed intervention (GEOMETRY ABSENT — must be designed)

WHAT WOULD MAKE US KILL IT?

killer experiment fails: falsification_test_from_candidate (decision scores: EIG 0.4557178723356231, EIG-per-cost 0.4557178723356231; the falsification test carries the pre-registered pass/fail before any transfer)

WHAT TRANSACTION COULD MAKE SENSE?

Sponsored Validation

What Exists Today

Existing:

- + Documented mechanism with governing equations
- + Computational model
- + 3 external evidence sources
- + Design inputs and outputs
- + Failure analysis
- + 10-step engineering build plan
- + Transfer boundary definition

Does Not Yet Exist:

- Physical prototype
- Qualified manufacturing process
- Clinical data
- Verified IP ownership
- Regulatory clearance
- Commercial product

Buyer Can Commission:

- > Specific bench experiment (see build plan)
- > Prototype fabrication
- > Materials study
- > Manufacturing feasibility study
- > Regulatory pre-submission

Development / De-risking Ladder

CURRENT STATE: ENGINEERING_DEFINITION

NEXT ENGINEERING STEP: first physical article implementing the proposed intervention (GEOMETRY ABSENT — must be designed)

NEXT VALIDATION STEP: Bench prototype validation per build plan

NEXT REGULATORY STEP: Pre-submission to FDA for pathway classification

COMMERCIALIZATION GATE: Clinical validation evidence required

TRANSFER GATE: Transfer-ready when physical validation + regulatory pathway + IP verified

Engineering Investment Ladder

Next \$5K action: WP-01 - first physical article implementing the proposed intervention (GEOMETRY ABSENT — must be designed) - NOT ESTABLISHED (to be quoted)

Basis: Build plan WP. Scope: physical bench/clinical experiment — NOT simulation. Uncertainty: Effort estimate, not quoted.

Next \$25K action: WP-03 - article for VF-001 - NOT ESTABLISHED (to be quoted)

Basis: Build plan WP. Scope: Compare the P07 model's viscosity predictions against benchmark measurements over 100 injection cycles with and without the shot-to-shot.... Uncertainty: Effort estimate, not quoted.

Next \$100K action: WP-05 - article for VF-003 - NOT ESTABLISHED (to be quoted)

Basis: Build plan WP. Scope: bench flow-loop characterization across operating range. Uncertainty: Effort estimate, not quoted.

Top 5 Buyer Diligence Questions

Q1: Is the mechanism physically credible for the intended application?

The governing model includes 3 equations. Key assumptions: CSF approximated as Newtonian (valid at shunt flow rates); no wall compliance effects; transition criterion $Re \sim 2300$ is a MODEL_DERIVED engineering rule of thumb; $C_d \sim 0.6-0.65$ is an engineering prior, not a measurement; FLUID-001 (Hagen-Poiseuille (laminar pipe flow)) is applicable only under: Newtonian fluid, laminar regime, rigid circular lumen, fully developed flow.; FLUID-002 (Reynolds number) is applicable only under: circular section; single-phase flow.; FLUID-003 (Orifice flow (sharp-edged restriction)) is applicable only under: turbulent orifice flow; C_d treated as MODEL_DERIVED until measured.; Model input variables requiring sourced values before any numeric result: r (m), dP (Pa), μ (Pa.s), L (m), ρ (kg/m³), v (m/s), D (m), μ (Pa.s), C_d (-), A (m²), dP (Pa), ρ (kg/m³) — all UNKNOWN (no sourced value exists; Art. VIII).; Model output variables produced by the governing model: Q (m³/s), Re (-), Q (m³/s) — UNKNOWN until the inputs are sourced and the model is evaluated.; Critical parameters under specification (parameters with recorded units and epistemic status; no values invented): lumen inner diameter [r , m] value_status=UNKNOWN; lumen length [r , m] value_status=UNKNOWN; surface roughness [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN; operating pressure head [dP , Pa] value_status=UNKNOWN; flow rate range [Q , m³/s] value_status=UNKNOWN; viscosity of flowed medium [μ , Pa.s] value_status=UNKNOWN; occlusion growth tolerance [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN.. UNKNOWN: whether assumptions hold in vivo.

Q2: What evidence supports the core claim?

3 external evidence sources. All are external precedent, not physical validation of this specific invention.

Q3: What is the regulatory pathway?

UNKNOWN. Not yet determined. Requires pre-submission to FDA. No predicate identified for some packages.

Q4: Who owns the IP?

UNKNOWN. IP ownership not verified. Requires legal counsel engagement.

Q5: What would it cost to validate?

First experiment: first physical article implementing the proposed intervention (GEOMETRY ABSENT — must be designed). Estimated effort from build plan. Full validation cost: COST TO BE QUOTED.

Commercial Interest

TARGET APPLICATION: CSF shunt systems for hydrocephalus management

CURRENT ALTERNATIVE: Standard fixed-pressure or programmable CSF shunt valves

MECHANISM DIFFERENCE: Shot-to-shot compensation mechanism that accumulates and transfers learning results across consecutive shots.

POTENTIAL VALUE DRIVER: Addresses: Documented problem 'parameter estimation error' concerning viscosity model. User-submitted engineering problem (session input): "Can the P-07 model's viscosity parameter be corrected from measured flow? (model correction semantics)". [cod_optimade] retrieval status EMPTY — provider outcome recorded, not treated as absence (Art. XXI.3). [arxiv] 5 records retrieved live (2001-2025). Illustrative records: 'Modelling fluid deformable surfaces with an emphasis on biological interfaces'; 'Droplet break-up with negative momentum Fluid Dynamics Videos'. LIMITS STAMPED INLINE: literature context establishes the problem class, not event rates. Problem-existence basis: live records retrieved.

ADOPTION BARRIER: Requires physical validation, regulatory clearance, and manufacturing qualification

DEVELOPMENT BARRIER: No physical prototype exists; manufacturing process not established

Licensee Capability Fit

IDEAL BUYER: domain-appropriate technology buyer (to be identified by portfolio owner)

REQUIRED ENGINEERING CAPABILITY: Catheter design, fluid mechanics / domain-specific engineering

REQUIRED MANUFACTURING CAPABILITY: Medical tubing extrusion, injection molding, cleanroom assembly

REQUIRED REGULATORY CAPABILITY: FDA 510(k) or PMA submission experience

REQUIRED CAPITAL: Validation: \$50-200K. Full development: \$500K-2M (estimates, not quoted)

REQUIRED MARKET ACCESS: Neurosurgery market channels; hospital purchasing relationships

1. Technology Description

Domain: fluidics_hydraulic

Disciplines: fluid mechanics, mechanical engineering, biomedical engineering, materials science

Architecture: PROPOSED architecture (no physical system exists): flow path / lumen architecture; pressure regulation element; sensing / feedback element (if closed-loop); termination interfaces (patient / reservoir) — realizing: Implement a cycle-to-cycle accumulation of viscosity estimates derived from measured flow rates, storing them in a passive lookup table that updates the P07 model's viscosity parameter before each new prediction.

Subsystems:

?: flow path / lumen architecture - ?

?: pressure regulation element - ?

?: sensing / feedback element (if closed-loop) - ?

?: termination interfaces (patient / reservoir) - ?

2. Mechanism Architecture

Physical Changes: PROPOSED: Implement a cycle-to-cycle accumulation of viscosity estimates derived from measured flow rates, storing them in a passive lookup table that updates the P07 model's viscosity parameter before each new prediction.

Key Physics: Fluid mechanics / hydraulics governing relations proposed for this invention; SYMBOLIC ONLY until inputs are sourced

3. Governing Engineering Model

Summary: Fluid mechanics / hydraulics governing relations proposed for this invention; SYMBOLIC ONLY until inputs are sourced

Equations:

FLUID-001: Hagen-Poiseuille (laminar pipe flow). Expression: $Q = (\pi * r^4 * \Delta P) / (8 * \mu * L)$.

Applicability to this invention: APPLICABLE; holds under: Newtonian fluid, laminar regime, rigid circular lumen, fully developed flow. Assumptions: CSF approximated as Newtonian (valid at shunt flow rates); no wall compliance effects. Engaged variables: μ .

FLUID-002: Reynolds number. Expression: $Re = (\rho * v * D) / \mu$. Applicability to this invention: APPLICABLE; holds under: circular section; single-phase flow. Assumptions: transition criterion $Re \sim 2300$ is a MODEL_DERIVED engineering rule of thumb. Engaged variables: μ .

FLUID-003: Orifice flow (sharp-edged restriction). Expression: $Q = C_d * A * \sqrt{2 * \Delta P / \rho}$. Applicability to this invention: CONDITIONAL; holds under: turbulent orifice flow; C_d treated as MODEL_DERIVED until measured. Assumptions: $C_d \sim 0.6-0.65$ is an engineering prior, not a measurement. Engaged variables: NOT ESTABLISHED.

Assumptions:

- CSF approximated as Newtonian (valid at shunt flow rates)
- no wall compliance effects
- transition criterion $Re \sim 2300$ is a MODEL_DERIVED engineering rule of thumb
- $C_d \sim 0.6-0.65$ is an engineering prior, not a measurement
- FLUID-001 (Hagen-Poiseuille (laminar pipe flow)) is applicable only under: Newtonian fluid, laminar regime, rigid circular lumen, fully developed flow.
- FLUID-002 (Reynolds number) is applicable only under: circular section; single-phase flow.
- FLUID-003 (Orifice flow (sharp-edged restriction)) is applicable only under: turbulent orifice flow; C_d treated as MODEL_DERIVED until measured.

- Model input variables requiring sourced values before any numeric result: r (m), dP (Pa), μ (Pa.s), L (m), ρ (kg/m³), v (m/s), D (m), μ (Pa.s), C_d (-), A (m²), dP (Pa), ρ (kg/m³) — all UNKNOWN (no sourced value exists; Art. VIII).
- Model output variables produced by the governing model: Q (m³/s), Re (-), Q (m³/s) — UNKNOWN until the inputs are sourced and the model is evaluated.
- Critical parameters under specification (parameters with recorded units and epistemic status; no values invented): lumen inner diameter [r , m] value_status=UNKNOWN; lumen length [r , m] value_status=UNKNOWN; surface roughness [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN; operating pressure head [dP , Pa] value_status=UNKNOWN; flow rate range [Q , m³/s] value_status=UNKNOWN; viscosity of flowed medium [μ , Pa.s] value_status=UNKNOWN; occlusion growth tolerance [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN.

Boundary Conditions:

- to be established by the first bench characterization (not yet performed)

Failure Regimes:

- {'regime': 'parameter estimation error', 'basis': 'SOURCE_FACT (problem.json)'}
- {'regime': 'proximal occlusion by tissue ingrowth', 'basis': 'domain registry candidate from the engineering domain registry — applicability to THIS invention recorded below'}
- {'regime': 'lumen collapse under external pressure (kink)', 'basis': 'domain registry candidate from the engineering domain registry — applicability to THIS invention recorded below'}
- {'regime': 'encrustation / mineral deposition', 'basis': 'domain registry candidate from the engineering domain registry — applicability to THIS invention recorded below'}
- {'regime': 'regime transition to turbulence altering calibration', 'basis': 'domain registry candidate from the engineering domain registry — applicability to THIS invention recorded below'}
- {'regime': 'check-valve seat wear opening reverse leakage', 'basis': 'domain registry candidate from the engineering domain registry — applicability to THIS invention recorded below'}

4. Design Inputs

ID	Input	Value	Origin
DI-001	Stated constraint	maintain model accuracy within specified tolerance	SOURCE_FACT
DI-002	Mechanism driver (proposed)	Shot-to-shot compensation mechanism that accumulates and transfers learnings across consecutive shots.	MODEL_USER
DI-003	Expected effect target	Reduces parameter estimation error by continuously refining the viscosity parameter with historical flow data, keeping model within	MODEL_USER
DI-004	Domain design input pattern: required flow applicability: variable pressure [head]	reduced flow applicability: variable pressure [head]	ENGINEERING_PROPOSED
DI-005	Domain design input pattern: medium viscosity: categorical data tokens [viscosity]	medium viscosity: categorical data tokens [viscosity]	ENGINEERING_PROPOSED
DI-006	Domain design input pattern: anticipated application: NOT_ESTABLISHED (candidate pattern; confirm against engineering profile)	anticipated application: NOT_ESTABLISHED (candidate pattern; confirm against engineering profile)	ENGINEERING_PROPOSED
DI-007	Domain design input pattern: device lifetime applicability: NOT_ESTABLISHED (candidate pattern; confirm against engineering profile)	device lifetime applicability: NOT_ESTABLISHED (candidate pattern; confirm against engineering profile)	ENGINEERING_PROPOSED
DI-008	Regulatory design input: compliance candidate standards: ISO 13485 (QMS) and further regulatory standards (e.g. FDA CFR 812.60)	compliance candidate standards: ISO 13485 (QMS) and further regulatory standards (e.g. FDA CFR 812.60)	ENGINEERING_PROPOSED
DI-009	Custodied observation: Frequency of data acquisition: 100 Hz (range 10-1000 Hz)	frequency of data acquisition: 100 Hz (range 10-1000 Hz)	7eb0d660134ba5bbe1611b
DI-010	Custodied observation: Towards physics-based / empirical / digital origin: 40% (range 0-100%)	towards physics-based / empirical / digital origin: 40% (range 0-100%)	3172f116bbbb40d3c
DI-011	Custodied observation: Development of a control system for a hydraulic actuator: 100% (range 0-100%)	development of a control system for a hydraulic actuator: 100% (range 0-100%)	76107eba83f9b18cd78302
DI-012	Custodied observation: Physics-Guided Multi-Task Learning for a Sensing System: 50% (range 0-100%)	physics-guided multi-task learning for a sensing system: 50% (range 0-100%)	cd31c31ff53533bb4
DI-013	Custodied observation: Transfer learning for a predictive model: 75% (range 0-100%)	transfer learning for a predictive model: 75% (range 0-100%)	56a0556795b81798e12a6

5. Design Outputs

ID	Description	Status	Missing Inputs
DO-001	flow path / lumen architecture realizing Implement a cycle to cycle accumulation of flow rates, storing them in a database	CONCEPTUAL	details of cycle accumulation, details of storage
DO-002	pressure regulation element realizing Implement a cycle to cycle accumulation of pressure, storing them in a database	CONCEPTUAL	details of cycle accumulation, details of storage
DO-003	sensing / feedback element (if closed-loop) realizing Implement a cycle to cycle accumulation of sensing data, storing them in a database	CONCEPTUAL	details of cycle accumulation, details of storage
DO-004	termination interfaces (patient / reservoir) realizing Implement a cycle to cycle accumulation of termination data, storing them in a database	CONCEPTUAL	details of cycle accumulation, details of storage
DO-005	flow path / lumen architecture interfaces with pressure regulation element: signal transfer / data exchange (if needed) to cycle accumulation	CONCEPTUAL	details of signal transfer, details of data exchange
DO-006	pressure regulation element interfaces with sensing / feedback element (if closed-loop): signal transfer / data exchange (if needed) to cycle accumulation	CONCEPTUAL	details of signal transfer, details of data exchange
DO-007	sensing / feedback element (if closed-loop) interfaces with termination interfaces: signal transfer / data exchange (if needed) to cycle accumulation	CONCEPTUAL	details of signal transfer, details of data exchange
DO-008	Device boundaries: patient/fluid inlet and outlet termination control/data processing to cycle accumulation of viscosity	CONCEPTUAL	details of device boundaries, details of data processing
DO-009	Flow/force/signal path geometry realizing the mechanism to cycle accumulation of flow/force/signal path geometry	UNKNOWN	details of flow/force/signal path geometry, details of cycle accumulation
DO-010	Operating range for lumen inner diameter within which the mechanism holds its stated behavior	UNKNOWN	details of operating range, details of mechanism behavior
DO-011	Operating range for lumen length within which the mechanism holds its stated behavior	UNKNOWN	details of operating range, details of mechanism behavior
DO-012	Operating range for surface roughness within which the mechanism holds its stated behavior	UNKNOWN	details of operating range, details of mechanism behavior
DO-013	Operating range for operating pressure head within which the mechanism holds its stated behavior	UNKNOWN	details of operating range, details of mechanism behavior
DO-014	Operating range for flow rate range within which the mechanism holds its stated behavior	UNKNOWN	details of operating range, details of mechanism behavior
DO-015	Operating range for viscosity of flowed medium within which the mechanism holds its stated behavior	UNKNOWN	details of operating range, details of mechanism behavior
DO-016	No closed-loop control is proposed: the invention operates as open-loop as specified; this record exists so the omission is explicit and auditable	CONCEPTUAL	
DO-017	Information pipeline: ingestion -> feature/representation -> UQ -> decision input/output to the loop, later stage of decision making, post-stage	PROPOSED	details of information pipeline, details of decision making
DO-018	Fixture enabling: bench flow-loop characterization across the operating range — rig characterization, build specification, acceptance instrumentation	UNKNOWN	details of fixture enabling, details of bench flow-loop characterization
DO-019	Fixture enabling: pressure-drop mapping vs flow (identifying transitions) — rig characterization, build specification, acceptance instrumentation	UNKNOWN	details of fixture enabling, details of pressure-drop mapping

ID	Description	Status	Missing Inputs
DO-020	Fixture enabling: accelerated occlusion / fouling challenge	UNKNOWN — rig concept finalized, build type, instrumentation	fixed geometry, fixture run time, occlusion; acceptance instrumentation
DO-021	Fixture enabling: pressure-flow mapping across full operating envelope	UNKNOWN — rig concept generated, build type, instrumentation	fixed geometry, fixture run time, occlusion; acceptance instrumentation

6. Critical Design Parameters

Name	Value	Unit	Basis	Verification
lumen inner diameter	UNKNOWN (no sourced value)	m	ENGINEERING_PROPOSED	(design assigned candidate) value and verification method
lumen length	UNKNOWN (no sourced value)	m	ENGINEERING_PROPOSED	(design assigned candidate) value and verification method
surface roughness	UNKNOWN (no sourced value)	UNKNOWN	ENGINEERING_PROPOSED	(design assigned candidate) value and verification method
operating pressure head	UNKNOWN (no sourced value)	Pa	ENGINEERING_PROPOSED	(design assigned candidate) value and verification method
flow rate range	UNKNOWN (no sourced value)	m^3/s	ENGINEERING_PROPOSED	(design assigned candidate) value and verification method
viscosity of flowed medium	UNKNOWN (no sourced value)	Pa.s	ENGINEERING_PROPOSED	(design assigned candidate) value and verification method
occlusion growth tolerance	UNKNOWN (no sourced value)	UNKNOWN	ENGINEERING_PROPOSED	(design assigned candidate) value and verification method

7. Failure Modes

Mode	Mechanism	Mitigation	Verification	Residual Uncertainty
parameter estimation	Err by parameter estimation	design check via FMEA, FTA, FISH-BONE, etc. the design	UNKNOWN	When the operational data and failure rate (Ae.p)
proximal occlusion by tissue ingrowth	Proximal occlusion by tissue ingrowth	by engineering with self-polishing surface and the growing distance	UNKNOWN	When the clinical assessment and failure rate (Ae.p)
lumen collapse under external pressure	Failure by pressure (risk) under external pressure (risk)	external pressure (risk) evaluation pressure bending shape	UNKNOWN	When the mechanical analysis and failure rate (Ae.p)
encrustation / mineral deposition	Failure by encrustation / mineral deposition	material selection, coating, etc.	UNKNOWN	When the material selection and failure rate (Ae.p)
regime transition to turbulence	Failure by regime transition to turbulence	building a large diameter / large margin	UNKNOWN	When the proposed condition and failure rate (Ae.p)
check-valve seat wear	Failure by check-valve seat wear	material selection, etc.	UNKNOWN	When the proposed condition and failure rate (Ae.p)

8. Failure Analysis

Failure Mode	Mechanism	Mitigation	Residual Uncertainty
parameter estimation error	Fails by parameter estimation error: the design failure rate is given as 1.1×10^{-5} per hour, which is based on a failure mode state (A in the V)	design failure rate is given as 1.1×10^{-5} per hour, which is based on a failure mode state (A in the V)	design failure rate is given as 1.1×10^{-5} per hour, which is based on a failure mode state (A in the V)
proximal occlusion by tissue ingrowth	Fails by proximal occlusion by tissue ingrowth: the cells proliferate and fill the catheter lumen, leading to proximal occlusion and subsequent failure of the catheter (A in the V)	proximal occlusion by tissue ingrowth: the cells proliferate and fill the catheter lumen, leading to proximal occlusion and subsequent failure of the catheter (A in the V)	proximal occlusion by tissue ingrowth: the cells proliferate and fill the catheter lumen, leading to proximal occlusion and subsequent failure of the catheter (A in the V)
lumen collapse under external pressure	Fails by lumen collapse under external pressure: the catheter lumen collapses under external pressure, leading to occlusion and subsequent failure of the catheter (A in the V)	lumen collapse under external pressure: the catheter lumen collapses under external pressure, leading to occlusion and subsequent failure of the catheter (A in the V)	lumen collapse under external pressure: the catheter lumen collapses under external pressure, leading to occlusion and subsequent failure of the catheter (A in the V)
encrustation / mineral deposition	Fails by encrustation / mineral deposition: mineral deposits form on the catheter lumen, leading to occlusion and subsequent failure of the catheter (A in the V)	encrustation / mineral deposition: mineral deposits form on the catheter lumen, leading to occlusion and subsequent failure of the catheter (A in the V)	encrustation / mineral deposition: mineral deposits form on the catheter lumen, leading to occlusion and subsequent failure of the catheter (A in the V)
regime transition to turbulence	Fails by regime transition to turbulence: the flow regime transitions from laminar to turbulent, leading to increased wear and subsequent failure of the catheter (A in the V)	regime transition to turbulence: the flow regime transitions from laminar to turbulent, leading to increased wear and subsequent failure of the catheter (A in the V)	regime transition to turbulence: the flow regime transitions from laminar to turbulent, leading to increased wear and subsequent failure of the catheter (A in the V)
check-valve seat wear opening	Fails by check-valve seat wear opening: the check-valve seat wears, leading to leakage and subsequent failure of the catheter (A in the V)	check-valve seat wear opening: the check-valve seat wears, leading to leakage and subsequent failure of the catheter (A in the V)	check-valve seat wear opening: the check-valve seat wears, leading to leakage and subsequent failure of the catheter (A in the V)

9. Verification Strategy

[illegible]

10. Validation Strategy

ID	Requirement	Method	Acceptance	Result
VA-001	end-use validation of verified design	NOT ESTABLISHED (defined after verification passes)	criteria to be fixed at NOT_DONE	NOT_DONE (no definition; validation is

11. Materials

Component	Candidate Material	Source	Verification Required	Status
domain candidate material	medical-grade silicone elastomer	standard CSF shunt catheter material	True	ENGINEERING_PROPOSED
domain candidate material	polycarbonate / PSU rigid components	standard shunt valve housings	True	ENGINEERING_PROPOSED

12. Bill of Materials

Item	Description	Qty	Type	Material	Supplier	Criticality	Verification
1	medical-grade silicone elastomer	1	Material	Medical-grade silicone elastomer	Not Established	Not Established	Applicability verification required
2	polycarbonate / PSU rigid components	1	Material	polycarbonate / PSU rigid components	Not Established	Not Established	Applicability verification required

13. Manufacturing

Status: NO MANUFACTURING ANALYSIS POSSIBLE — no design exists; processes are candidates only

Candidate Processes:

- medical extrusion with inner-diameter tolerance control: domain registry manufacturing route - candidate only; no process qualification exists
- injection molding of valve/housing features: domain registry manufacturing route - candidate only; no process qualification exists
- laser drilling of calibrated orifices: domain registry manufacturing route - candidate only; no process qualification exists

14. External Evidence

Source 1:

URL :

Excerpt:

Source 2:

URL :

Excerpt:

Source 3:

URL :

Excerpt:

15. Transfer Boundary

YOU RECEIVE:

- + Buyer receives: canonical invention specification (with epistemic classes).
- + Buyer receives: engineering specification (this document, all blocks classed).
- + Buyer receives: design input/output graph with explicit IDs.
- + Buyer receives: engineering reasoning chains (claim->principle->model->input->assumption->output->failure->verification).
- + Buyer receives: killer-experiment definition and its EIG basis.
- + Buyer receives: evidence custody references and hashes.
- + Buyer receives: unresolved-uncertainty register.
- + Buyer receives: domain equation set (symbolic; sources and applicability judgments identified).

YOU MUST DEVELOP:

- > Buyer must create: actual geometry and detailed design.
- > Buyer must create: sourced engineering parameters and acceptance thresholds.
- > Buyer must create: physical prototypes.
- > Buyer must create: all verification and validation results.
- > Buyer must create: manufacturing process qualification.
- > Buyer must create: regulatory pathway determination and submissions.

YOU MUST VERIFY:

- ? Biocompatibility (ISO 10993)
- ? Sterilization compatibility
- ? Mechanical integrity
- ? Regulatory pathway
- ? IP ownership and freedom to operate
- ? Manufacturing process capability

16. Open Questions

- {'unknown': 'FM_WITHOUT_QUANTITY_MATCHED_VERIFICATION:FM-DOM-003', 'reason': 'no available verification method measures a quantity family this failure affects (E21-A quantity rule); UNKNOWN disclosure recorded on the row'}
- {'unknown': 'value of critical parameter lumen inner diameter [r m]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-001'}
- {'unknown': 'value of critical parameter lumen length [r m]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-002'}
- {'unknown': 'value of critical parameter surface roughness [NOT_ASSIGNED UNKNOWN]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-003'}
- {'unknown': 'value of critical parameter operating pressure head [dP Pa]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-004'}
- {'unknown': 'value of critical parameter flow rate range [Q m^3/s]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-005'}
- {'unknown': 'value of critical parameter viscosity of flowed medium [mu Pa.s]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-006'}
- {'unknown': 'value of critical parameter occlusion growth tolerance [NOT_ASSIGNED UNKNOWN]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-007'}
- {'unknown': 'acceptance criterion for verification VF-001', 'reason': 'criterion status ENGINEERING_PROPOSED: no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline', 'binding': 'VF-001'}

- {'unknown': 'acceptance criterion for verification VF-002', 'reason': 'criterion status ENGINEERING_PROPOSED: no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline', 'binding': 'VF-002'}
- {'unknown': 'acceptance criterion for verification VF-003', 'reason': 'criterion status ENGINEERING_PROPOSED: no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline', 'binding': 'VF-003'}
- {'unknown': 'acceptance criterion for verification VF-004', 'reason': 'criterion status ENGINEERING_PROPOSED: no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline', 'binding': 'VF-004'}
- {'unknown': 'acceptance criterion for verification VF-005', 'reason': 'criterion status ENGINEERING_PROPOSED: no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline', 'binding': 'VF-005'}
- {'unknown': 'acceptance criterion for verification VF-006', 'reason': 'criterion status ENGINEERING_PROPOSED: no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline', 'binding': 'VF-006'}
- {'unknown': 'acceptance criterion for verification VF-007', 'reason': 'criterion status ENGINEERING_PROPOSED: no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline', 'binding': 'VF-007'}
- {'unknown': 'acceptance criterion for verification VF-008', 'reason': 'criterion status ENGINEERING_PROPOSED: no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline', 'binding': 'VF-008'}

17. Decisive Next Experiment

WP	Test Article	Equipment	Measurement	Acceptance	Deliverable	Effort
WP-01	first physical article	implementing the proposed physical verification (0.5 GPa) test	0.5 GPa test	ENGINEERING_PROPOSED: no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline	0.5 GPa test	0.5 GPa test
WP-02	constraint-compliance envelope article	per method	constraint-compliance envelope article	ENGINEERING_PROPOSED: no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline	constraint-compliance envelope article	constraint-compliance envelope article
WP-03	article for VF-001	per method	Compare the P107 model to the actual flow	ENGINEERING_PROPOSED: no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline	Compare the P107 model to the actual flow	Compare the P107 model to the actual flow
WP-04	article for VF-002	per method	run killer experiment: NOISE TEST	ENGINEERING_PROPOSED: no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline	run killer experiment: NOISE TEST	run killer experiment: NOISE TEST
WP-05	article for VF-003	per method	bench flow-loop characterization	ENGINEERING_PROPOSED: no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline	bench flow-loop characterization	bench flow-loop characterization
WP-06	article for VF-004	per method	pressure-drop mapping	ENGINEERING_PROPOSED: no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline	pressure-drop mapping	pressure-drop mapping
WP-07	article for VF-005	per method	delivered-flow trend	ENGINEERING_PROPOSED: no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline	delivered-flow trend	delivered-flow trend
WP-08	article for VF-006	per method	kink/collapse test across 1000 ft	ENGINEERING_PROPOSED: no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline	kink/collapse test across 1000 ft	kink/collapse test across 1000 ft
WP-09	article for VF-007	per method	pressure-flow mapping	ENGINEERING_PROPOSED: no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline	pressure-flow mapping	pressure-flow mapping
WP-10	article for VF-008	per method	reverse-leakage rate	ENGINEERING_PROPOSED: no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline	reverse-leakage rate	reverse-leakage rate

18. Buyer Decision Framework

WHY INVEST: Engineering definition exists with 3 governing equations and a 10-step build plan. The next experiment is clearly specified. Target buyer: domain-appropriate technology buyer (to be identified by portfolio owner).

WHY WAIT: No physical prototype exists. Regulatory pathway is unresolved. IP ownership is unverified. Manufacturing process is not qualified.

WHY REJECT: killer experiment fails: falsification_test_from_candidate (decision scores: EIG 0.4557178723356231, EIG-per-cost 0.4557178723356231; the falsification test carries the pre-registered pass/fail before any transfer)

WHAT WOULD CHANGE THE DECISION: A successful bench experiment demonstrating the core mechanism. A qualified regulatory pathway. A confirmed IP position.

19. Kill Condition

KILL_IF: killer experiment fails: falsification_test_from_candidate (decision scores: EIG 0.4557178723356231, EIG-per-cost 0.4557178723356231; the falsification test carries the pre-registered pass/fail before any transfer)

DISCLOSURE

This dossier distinguishes established evidence, external precedent, computational/modelled results, engineering proposals, and unresolved questions. The absence of physical validation should not be interpreted as evidence that the technology will fail; it means the relevant proposition has not yet been established experimentally.

This dossier does not constitute legal, patentability, FTO, regulatory, or investment advice.